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Chapter 3

Result And Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.

The first stage of the optimization process consist of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the AOM amplitude, and the compensation coil currents.

Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

3.1 Molasses temperature optimization

Before performing any optimization, the six beams in the vacuum chamber were realigned in order to minimize their overlap mismatch at the MOT position. This procedure was necessary since thermal drifts can induce misalignment in the optical components.

Preliminary values for the compensation coil currents have also been obtained prior the realignment using the Release-and -Recapture (R&R) technique:

$$\begin{aligned} I_x &= 80 \text{ mA} \\ I_y &= -200 \text{ mA} \\ I_z &= -400 \text{ mA}. \end{aligned} \tag{3.1}$$

These values were used as the starting point for the subsequent optimization procedure.

3.1.1 Detuning frequency optimization

The first optimization step consisted of measuring the molasses temperature for different values of the cooling laser detuning. The detuning was varied from -24.4 MHz ($4.0 \Gamma/2\pi$)

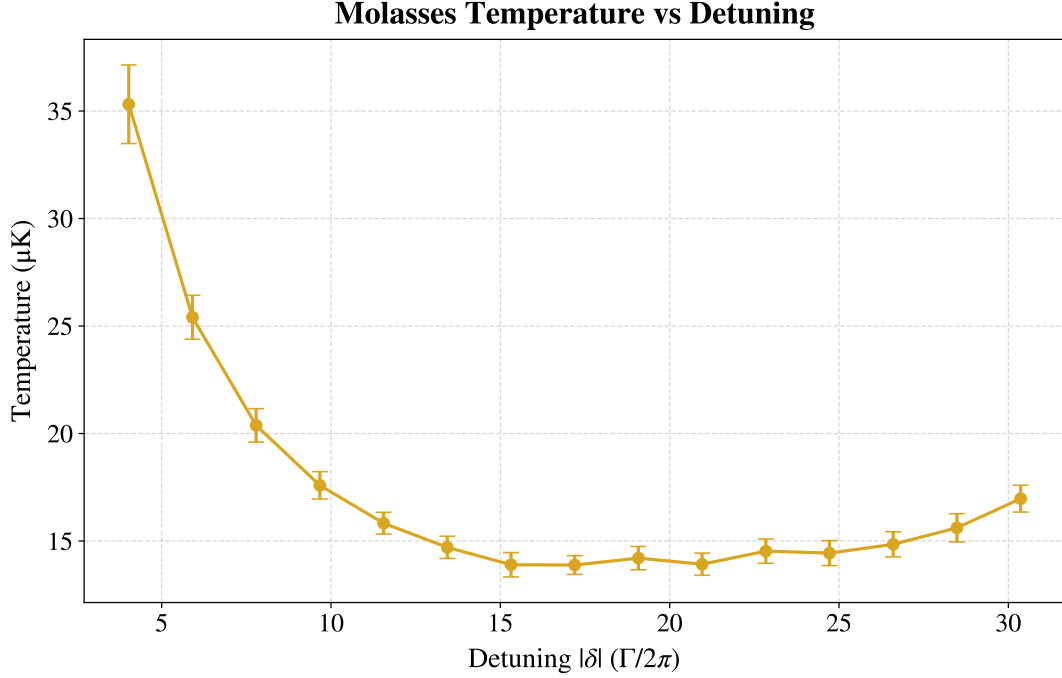


Figure 3.1: *Optical molasses temperature as a function of laser detuning. The horizontal axis represents the absolute value of the applied red detuning $|\delta|$, normalized to the natural linewidth $\Gamma/2\pi$. The data exhibits optimal sub-Doppler cooling in the region $|\delta| \approx 15 - 20$, reaching a minimum temperature of $13.9 \pm 0.4 \mu\text{K}$. For detunings up to $|\delta| \approx 15$, the temperature follows the $1/|\delta|$ dependence expected for an ideal Sisyphus cooling. At larger detunings, the curve flattens and eventually begins to rise.*

to -184 MHz ($30.4 \Gamma/2\pi$), with a step of -11.4 MHz .

The resulting average temperature is shown in Figure 3.1 as a function of the detuning expressed in units of $\Gamma/2\pi$. At small detunings, the data follow the expected $1/|\delta|$ dependence typical of the Sisyphus cooling. As the detuning increases two phenomena become relevant: excessive decrease of the optical cycle rate $\Gamma_p \propto 1/\delta^2$ and the approach of the transition $F = 2 \rightarrow F' = 2$ which is found 267 MHz below the cooler target transition $F = 2 \rightarrow F' = 3$ (see Figure ref??). These effects cause the temperature curve to flatten in the region $|\delta| \approx 15 - 20$ and eventually start to increase again for larger detunings.

The lowest average temperature was obtained for a detuning of $|\delta| = 17.2$ corresponding to $|\delta| = 104.2 \text{ MHz}$. Figure 3.2 shows the fit of equation (2.9) for the optimal temperature configuration. The resulting temperature along the Cartesian axis are $T_x = (14.1 \pm 0.5) \mu\text{K}$ and $T_y = (13.5 \pm 0.7) \mu\text{K}$. Their weighted average yields the final temperature $T = (13.9 \pm 0.4) \mu\text{K}$.

As observed in Figure 3.2, the data points at long TOF diverge appreciably from the linear trend, consistently falling below the expected values. This behavior is consistently observed for all measurements. The hypothesis is that at sufficiently long TOF the cloud radius becomes non-negligible compared to the probe beam radius, violating the assumption of uniform illumination. This makes the outer region of the cloud lie in a region

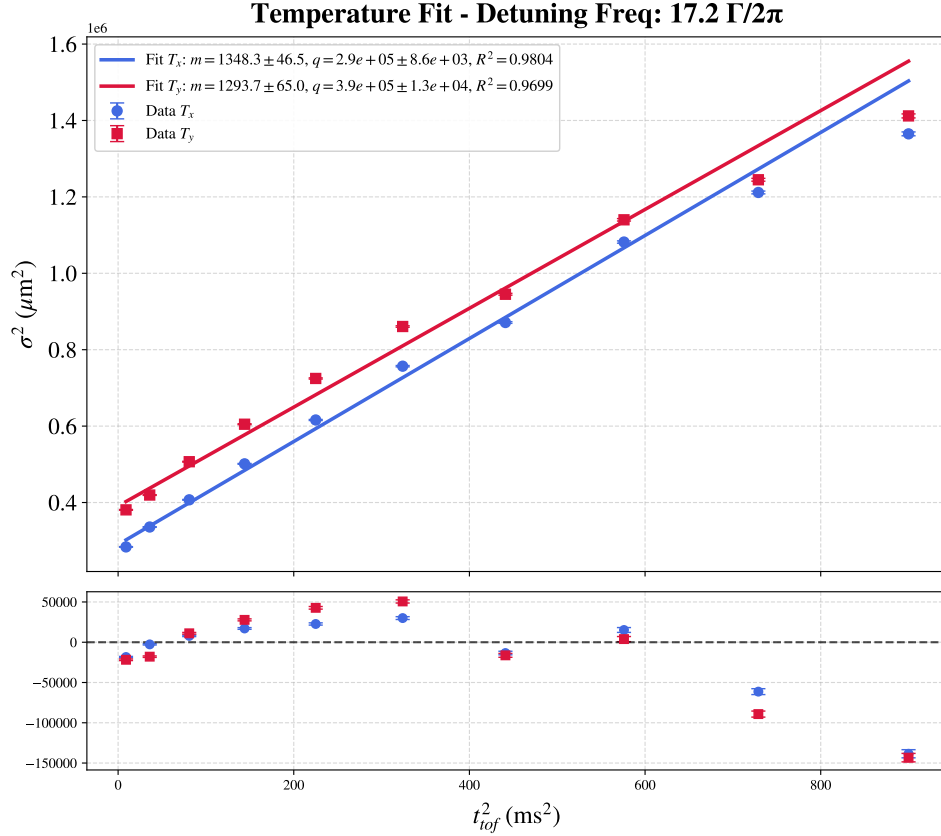


Figure 3.2: *Optical molasses temperature fit for $|\delta| = 17.2 \Gamma/2\pi$ with residuals. The large residuals of the last three data points from the linear fit are attributed to the cloud expansion, which makes its radial size non-negligible compared to the probe beam profile.*

of lower intensity, which leads the Gaussian fit to systematically underestimate the true variance. Fortunately, this effect is expected to become less disturbing as the temperature is lowered.

3.1.2 AOM amplitude optimization

As a second optimization step, the influence of cooling beam power was investigated by varying the AOM drive amplitude and the duration of the amplitude ramp. The ramp is applied at the beginning of the molasses phase. The AOM amplitude expressed in arbitrary units (a.u.) controls the RF amplitude driving the AOM. Three values were tested: 385, 475 and 550 a.u., corresponding respectively to approximately 0.7, 0.85 and 1.0 times the AOM amplitude during the MOT phase. In addition, ramp durations of 5 ms and 20 ms were tested for the transition of the AOM amplitude from the MOT value to the molasses value. The measurements were repeated for different values of the cooling detuning. The results of the optimization are shown in figure 3.3. Although small variations in the expected values of the temperature can be observed for different values of the AOM amplitude, these differences are smaller compared to the errors bars. There-

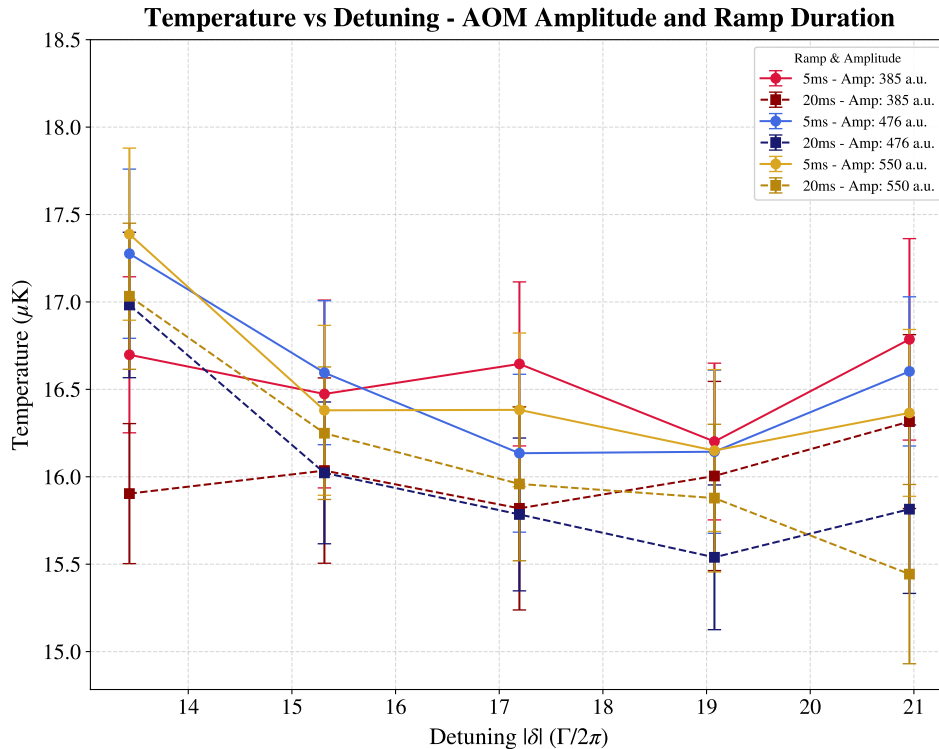


Figure 3.3: *Optical molasses temperature versus cooler detuning for different AOM amplitude and ramp durations. No clear optimal AOM amplitude is observed as the variations remain within the uncertainty. Temperatures corresponding to a ramp duration of 20 ms (dashed lines) are systematically lower than those obtained with 5 ms ramp (solid lines).*

fore, the measurements do not show statistically significant dependence of the molasses temperature on the AOM amplitude, and no clear optimum could be identified within the tested range. However, lower amplitudes were not investigated to avoid operating the AOM too far from its optimal efficiency regime.

On the other hand, as shown in Figure 3.3, the temperatures obtained with a 20ms ramp are systematically lower than those obtained with a 5 ms ramp, yielding an average temperature reduction of approximately $0.5 \mu\text{K}$. This improvement was relatively small, and was considered sufficient for the scope of the experiment, since further optimization of the order of few tens of microkelvin was not required, no other ramp times were tested. Consequently, a ramp duration of 20 ms was selected for the subsequent measurements.

It is worth noting that during the ODT loading characterization, this optimization was found to introduce additional fluctuations in the system. For this reason, the subsequent optimization of the compensation coil currents was carried out using an AOM amplitude of 385 a.u. corresponding to the lowest temperature measured at $|\delta| = 17.2 \Gamma/2\pi$, with a ramp duration of 20 ms. However, during the later optimization of the ODT, these parameters were adjusted again. Nevertheless, the measurements presented in this section revealed that their influence on the molasses temperature remains limited to a few tens of microkelvin, and therefore does not significantly affect the final temperature achieved.

3.1.3 Compensation field optimization

The final stage of the molasses optimization procedure, consisted in adjusting the compensation magnetic field. The compensation coil currents are set to the molasses values 5 ms after the MOT magnetic field had been switched off.

The optimization was performed sequentially. First, the z-axis current was optimized while keeping the x- and y-axis currents fixed to the preliminary values (3.1). Then the x-axis current was optimized using the optimal z value, and finally the y-axis current was optimized using the optimal currents for both the z- and x-axis. The results are shown in figures 3.4, 3.5 and 3.6 for the z-, x-, and y-axis coils respectively.

The resulting optimized currents used in the subsequent ODT loading stage are:

$$\begin{aligned} I_x &= -15 \text{ mA} \\ I_y &= -185 \text{ mA} \\ I_z &= -230 \text{ mA} \end{aligned} \tag{3.2}$$

The value of the z-axis compensation coil current deviated the most from its preliminary value, while the x-axis current underwent a smaller but still appreciable correction. In contrast, the y-axis coils current remained close to its preliminary values.

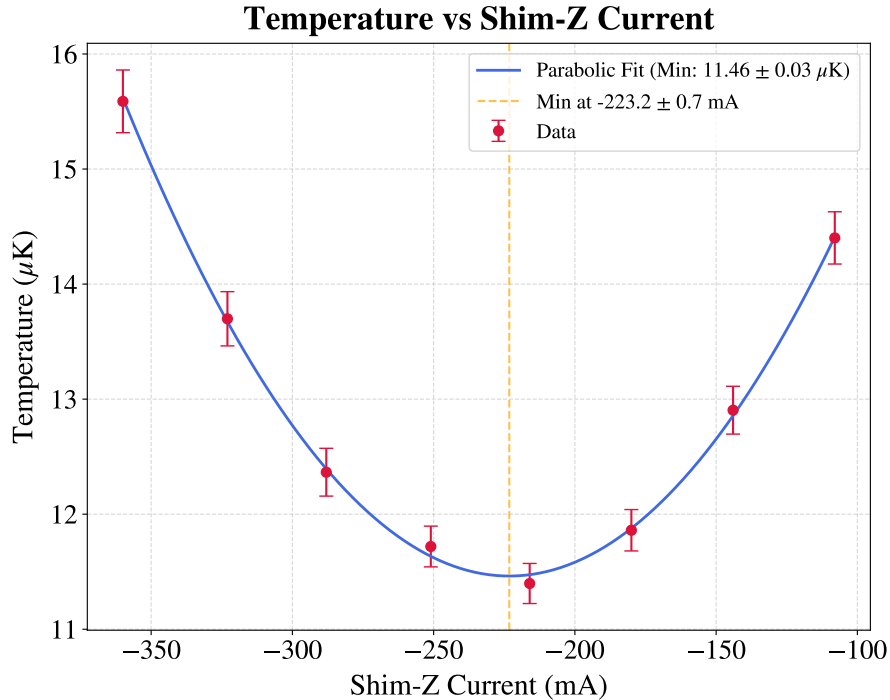


Figure 3.4: *Optical molasses temperature versus. Shim-Z current. The parabolic fit retrieves a minimum temperature $T = (11.46 \pm 0.03) \mu\text{K}$ at a current $I_z = (-223.2 \pm 0.7) \text{ mA}$.*

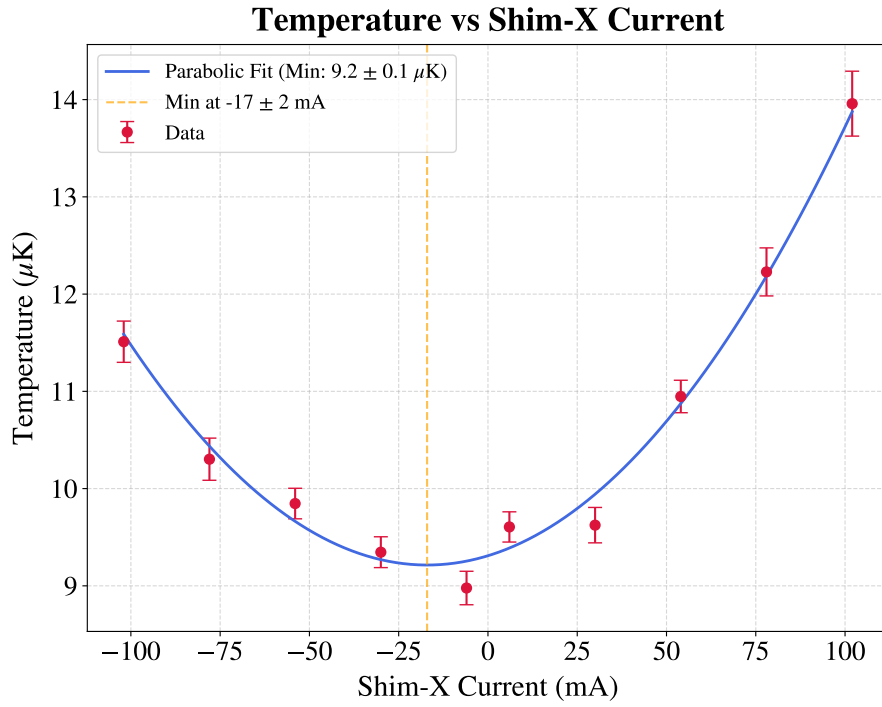


Figure 3.5: *Optical molasses temperature versus Shim-X current. The parabolic fit retrieves a minimum temperature $T = (9.2 \pm 0.1) \mu\text{K}$ at a current $I_x = (-17 \pm 2) \text{ mA}$.*

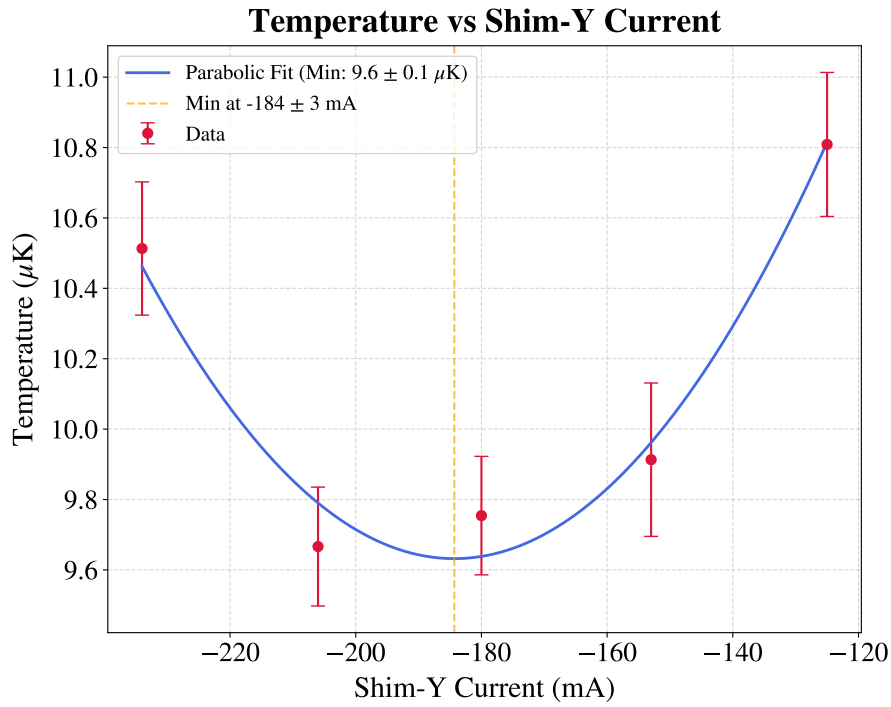


Figure 3.6: *Optical molasses temperature versus Shim-Y current. The parabolic fit retrieves a minimum temperature of $T = (9.6 \pm 0.1) \mu\text{K}$ at a current $I_y = (-184 \pm 3) \text{ mA}$.*

3.2 ODT loading optimization

The optimal parameters obtained for the molasses were used for the optimization of the ODT loading. This optimization was mainly focussed on the compensation magnetic field. The idea is to load the MOT and, before switching to the molasses phase, exploit the compensation magnetic field to shift the MOT position.

The MOT is formed where the magnetic field is approximately zero, thus by adjusting the compensation coil currents, it is possible to shift the zero and move the MOT. This makes the MOT loose atom since it is moved away from the optimal position of the intersection of the beams, but there is a trade-off with the better intersection with the ODT beam, that might increase the number of captured atoms. As long as this shift is small the molasses optimal parameters found in the previous sections of this chapter should nullify the residual magnetic field well, and they do not need a second revision.

In particular the compensation coil currents are adjusted to the loading values after load with 200 ms ramp in 20 steps. The time and the ramp are needed to make the translation smooth, in order to minimize the losing of atom in the process.

As for the molasses, also for the ODT loading preliminary values obtained prior to re-alignment and molasses optimization were available:

$$\begin{aligned} I_x &= -200 \text{ mA} \\ I_y &= -350 \text{ mA} \\ I_z &= 50 \text{ mA}. \end{aligned} \tag{3.3}$$

3.2.1 Shifted MOT

3.2.1.1 Preliminary measurements

The first measurements were taken using the FLIR camera and by employing the optimal parameters from the molasses temperature optimization, and the preliminary values for the shim for the ODT loading. Using an analysis procedure similar to that described in Section 2.3.3, without the rescaling of the background, the data were analyzed to characterize the ODT capturing. The resulting intensity difference images are shown in Figure 3.7 (2D image) and in Figure 3.8 (1D flattened image). Upon inspection of the 2D image, no brighter area that would represent the atom captured by the ODT is recognizable. By inspecting the 1D image, instead, seem that some pixel (~ 4 pixel) about the fiber position present a higher intensity. Other measurements revealed a similar behavior, confirming that the brightness of these pixels is given by the trapped atom in the ODT. However, given the small amplitude of the signal, in many images, it was not distinguishable from the noise level. The flatten image are precisely meant to increase the

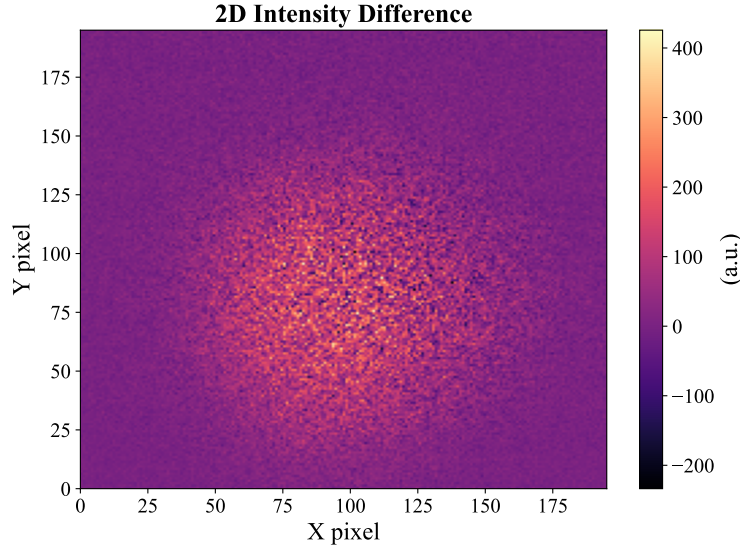


Figure 3.7: *Intensity difference 2D image, obtained with the FLIR camera and 20 ms of AOM amplitude ramp duration during the molasses phase. The intensity distribution does not show any brighter area that can be attributed to the presence of the ODT.*

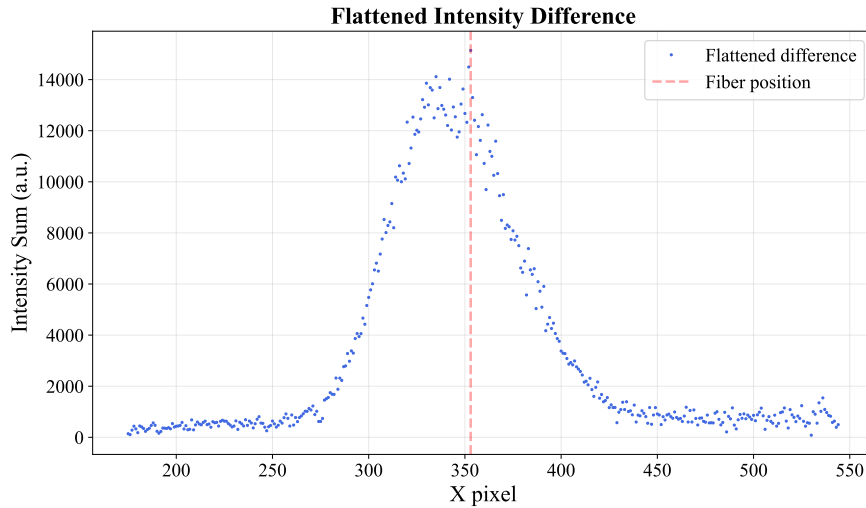


Figure 3.8: *Intensity difference 1D flattened image, obtained with the FLIR camera and 20 ms of AOM amplitude ramp duration during the molasses phase. The red vertical line gives an approximate position of the fiber center. Some higher intensity pixels are recognizable in correspondence of the ODT position, but clearly the averaging and difference are not effective in removing the background.*

signal-to-noise ratio thus having still a bad signal-to-noise ratio is not good. Furthermore, the presence of the residual pedestrian that is much higher with respect to the ODT signal is a clue that there are some oscillation noise in the system that are not averaged by the acquisition of five images, either because has a large standard deviation or a correlated behavior from one experiment to another. The larger concern was though the small ODT signal, which even if the system is supposed to be far from the optimal compensation current was expected to be higher. Another problematic arises, which is the narrowness

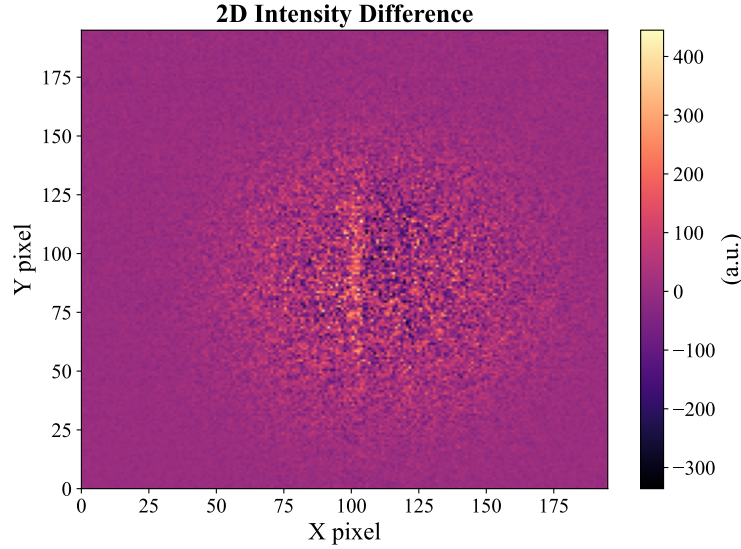


Figure 3.9: *Intensity difference 2D image, obtained with the FLIR camera and 5 ms of AOM amplitude ramp duration during the molasses phase. A brighter area is clearly visible in correspondence of the ODT.*

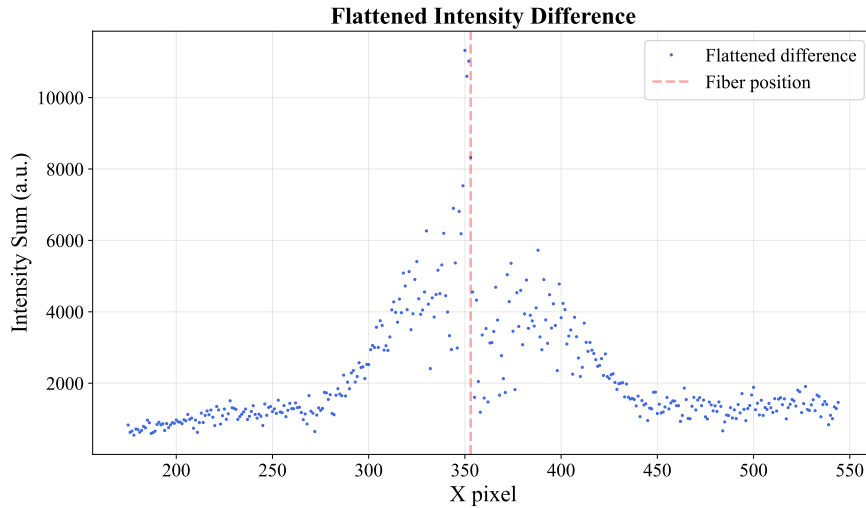


Figure 3.10: *Intensity difference 1D flattened image, obtained with the FLIR camera and 5 ms of AOM amplitude ramp during the molasses phase. The red vertical line gives an approximate position of the fiber center. With 5 ms ramp the averaging and the subtraction becomes more effective in separate the ODT signal from the background. The problem of the small number of pixel that correspond to the ODT signal is still present.*

of the area characterized by the ODT signal. Indeed, if the ODT signal occupies just few pixels the analysis is of particular difficulty because no Gaussian fit are possible, and also calculating the area as the sum of the pixel value carries a lot of uncertainty due to the poor resolution. This was caused by two factors, the difference of the image with the ODT on and images with ODT off have a residual background large compared with the expected signal of the ODT, therefore the signal is submerged in the noise, second the ODT beam radius is small compared with the cloud radius, therefore the ODT should be

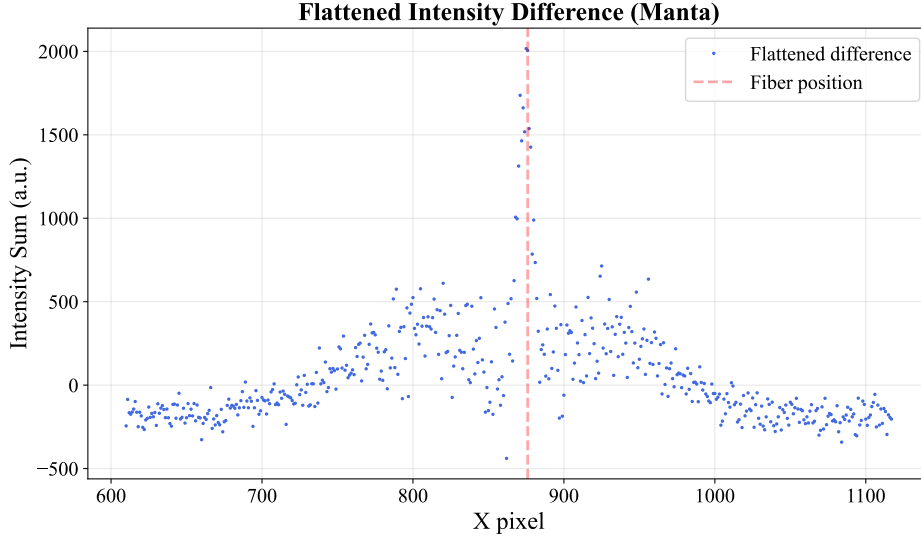


Figure 3.11: *Intensity difference 1D flattened image, obtained with the Manta camera and 5 ms of AOM amplitude ramp during the molasses phase. The combination of the lower pixel size of the Manta and the nearer position at which it was mounted increased the dimension in pixels of the ODT signal.*

visible just in on few number of pixel.

Since the preliminary analysis gave far better results, the parameter of the molasses were systematic changed to verify which caused the problematic. This study brought to the fact that the problematic was caused by the AOM ramp, during the molasses phase. In particular the main problem was caused by the ramp duration: switching from a 20 ms to a 5 ms ramp changed. Furthermore, to reduce as much as possible the fluctuation caused by the AOM switch, also the target amplitude was switched to 480 a.u., which represent the maximum amplitude but for the same reason the point width minimum derivative, thus the point in which the oscillation of the amplitude cause the lower oscillation on the actual power of the beam, stabilizing the system.

The results obtained with this configuration are shown in Figures 3.9 (2D difference) and 3.10 (1D difference). In the 2D plot a clear brighter strip is recognizable representing the atom captured in the ODT. By comparison of the 1D graphs in the Figures 3.8 and 3.10, it is evident how the pedestrian in the case of 5 ms ramp is smaller, having a maximum and an area of approximately half the case with 20 ms ramp. This is sufficient to make the ODT signal clearly emerge from the residual pedestrian and from the noise.

A problematic is still present which is the fact that with the magnification of the FLIR. For this reason, for the subsequent measurements the Manta camera, that has smaller pixels, was employed, by also mounting it as near as to exploit the minimum working distance of the MVL50M23 objective. By comparing the size of the fiber measured with the Manta and the FLIR, it is deduced that the apparent size of the object in pixels increases by a factor of ~ 2.25 . This scaling factor arises from the combined contribution of the higher

optical magnification and the smaller pixel size of the second imaging system.

3.2.1.2 Results

The systematic optimization of the compensation coil currents was carried out following the complete analysis procedure detailed in Section 2.3.3, utilizing the Manta camera. The procedure was performed sequentially along the spatial axes in the order $x \rightarrow z \rightarrow y$. During each scan, the currents of the remaining non-optimized coils were maintained at the preliminary values established in Equation (3.3).

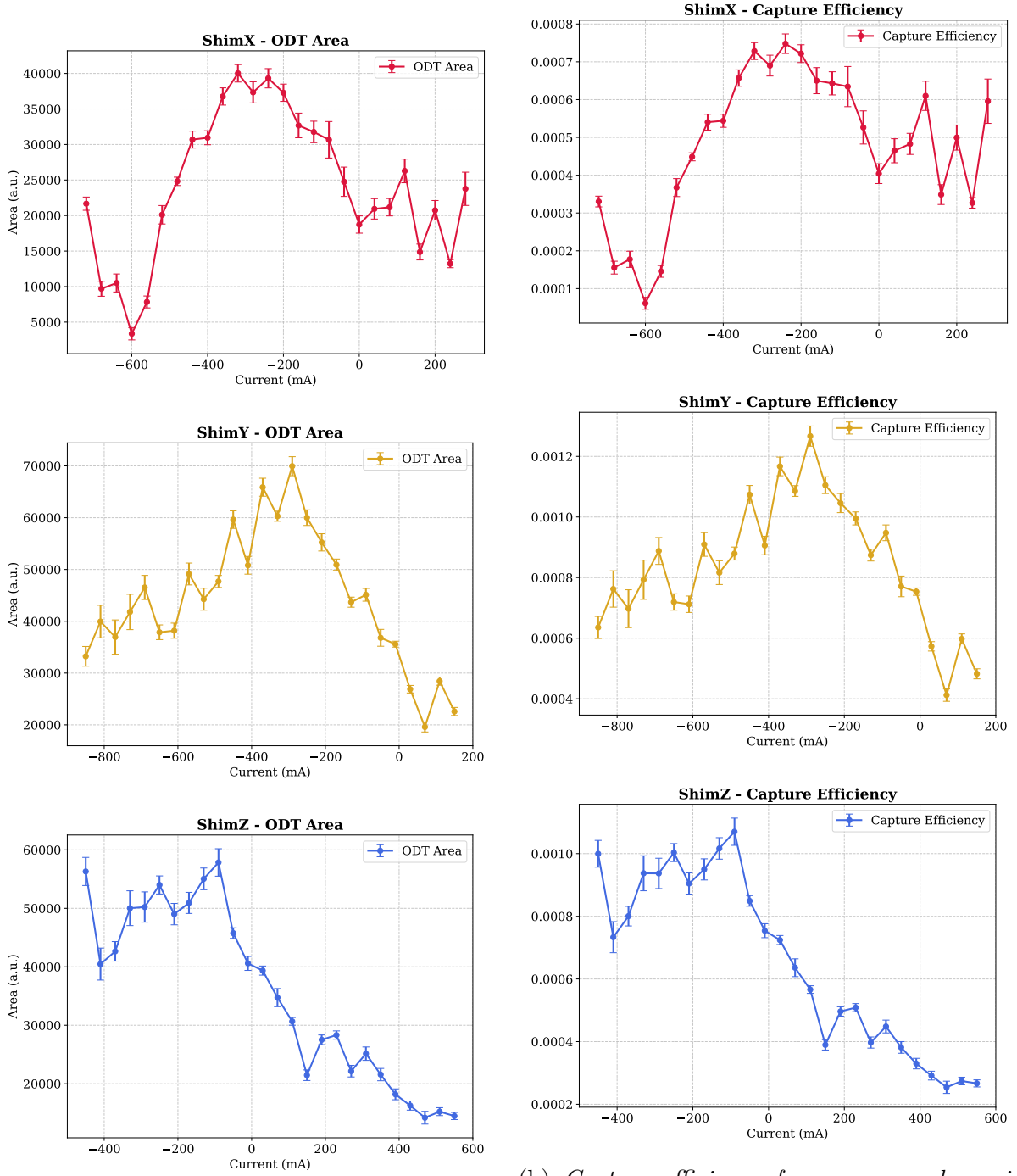
The capture efficiencies extracted from these scans are summarized in Figure 3.12b for the x -, y -, and z -coils. Similarly, Figure 3.12a displays the evaluated ODT signal area for each scan. This area is directly proportional to the total number of captured atoms, making it the physical quantity of fundamental interest. However, since the camera was not calibrated for absolute atom number measurements, this quantity can only be expressed in arbitrary units. Nonetheless, these values remain perfectly comparable provided the camera gain is kept constant. As evident from the plots, the behavior of the capture efficiency mirrors the trend of the ODT area.

This initial optimization yielded the following optimal currents:

$$\begin{aligned} I_x &= -220 \text{ mA}, \\ I_y &= -290 \text{ mA}, \\ I_z &= -90 \text{ mA}. \end{aligned} \tag{3.4}$$

To confirm the optimization achieved in the initial measurements, a second run was performed, yielding unconventional results regarding the z -axis compensation coil (Figure 3.13). As clearly shown in Figure 3.13a, the signal does not follow an expected parabolic or bell-shaped profile with a clear maximum. Instead, the ODT area and the capture efficiency exhibit an oscillatory behavior. Starting from a current of -1000 mA, the signal increases to a first local maximum at -800 mA, drops until ~ -700 mA, and then rises again to reach subsequent local maxima at ~ -600 mA and ~ -400 mA.

A similar periodic trend is observed in the MOT background area (Figure 3.13b), which displays peaks at ~ -800 mA, ~ -600 mA, and ~ -400 mA, matching the ODT area oscillation with good precision. While a proportional increase in the captured ODT fraction is naturally expected when the initial MOT atom number is higher, the oscillatory nature of this trend is peculiar. Tracking the x -centroid of the atomic cloud on the camera's image plane (Figure 3.13d) reveals an analogous spatial oscillation, a phenomenon not observed during the scans of the other two compensation axes. The likely physical explanation for this behavior lies in an imperfect spatial overlap of the cooling beams. As the z -axis compensation field is scanned, the position of the magnetic zero shifts along the laboratory z -axis as expected. However, a non-ideal beam intersection generates mul-



(a) ODT area for x-, y-, and z-axis scans.

(b) Capture efficiency for x-, y-, and z-axis scans.

Figure 3.12: ODT area (left) and capture efficiency (right) for the compensation coil current scans along x-, y-, and z-axis (performed in the order $x \rightarrow z \rightarrow y$). The qualitative agreement between the two observables is clearly visible for all the investigated axes.

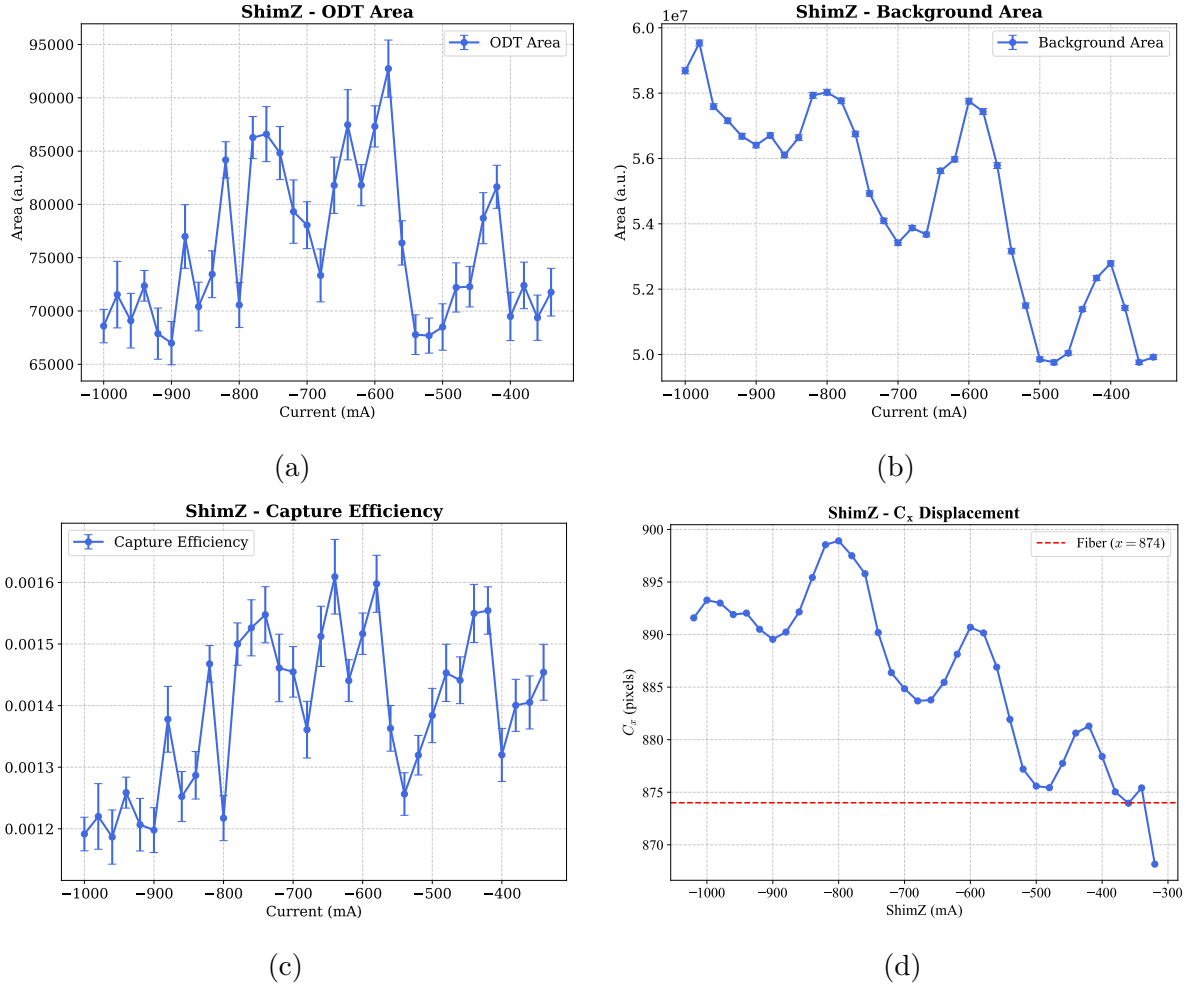


Figure 3.13: *Optimization results for the second set of measurements of the z -compensation coil current: (a) ODT signal area, (b) background area, (c) capture efficiency, (d) atomic cloud center position projected on the camera's x -axis.*

multiple localized regions (hotspots) of varying optical intensity. Since the MOT equilibrium position is determined by the balance between the Zeeman-induced magnetic force and the local radiation pressure, the atomic cloud is selectively pushed toward these intensity maxima. This results in a combined motion on the laboratory z - x plane, which projects as a spatial oscillation onto the 2D camera sensor. Furthermore, when the atoms traverse these optimal optical hotspots, the cooling process becomes locally more efficient. This leads to lower temperatures and significantly reduced atom losses, which translates to a higher overall atom number in the MOT (accounting for the peaks in the background signal) and, consequently, an enhanced capture efficiency into the dipole trap.

Finally, the optimal configuration maximizing the ODT loading was found to be:

$$\begin{aligned}
 I_x &= -220 \text{ mA}, \\
 I_y &= -280 \text{ mA}, \\
 I_z &= -600 \text{ mA}.
 \end{aligned}
 \tag{3.5}$$

This configuration yielded the largest ODT trap area, representing the maximum number of captured atoms, and is associated with a capture efficiency of $(0.160 \pm 0.005)\%$. The corresponding 2D difference image is shown in Figure 3.14.

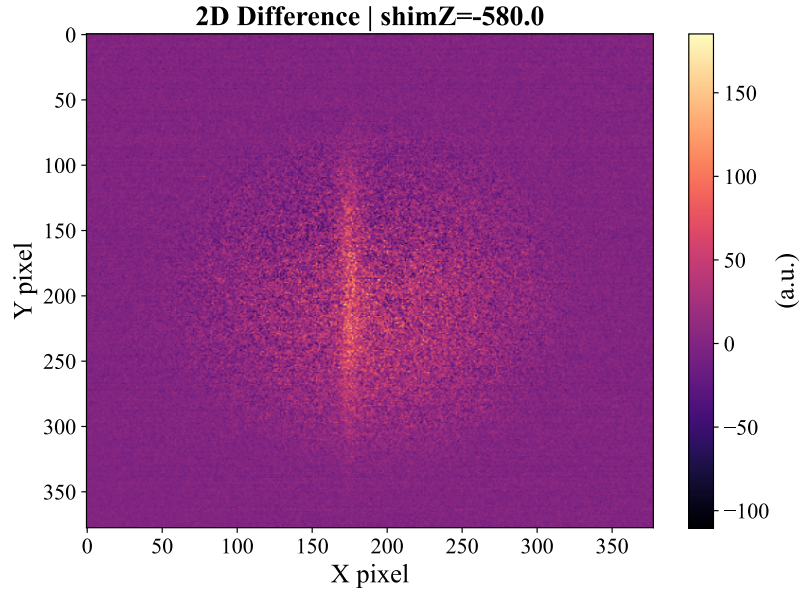


Figure 3.14: *Intensity difference 2D image obtained with the optimal currents: $I_x = -220$ mA, $I_y = -280$ mA, and $I_z = -600$ mA, corresponding to a capture efficiency of $\sim 0.16\%$.*